



Society of Petroleum Engineers

SPE-229785-MS

Application of Intelligent Waterflood Optimization for the Greater Burgan Oilfield

K. Alrasheedi, N. Alkandari, E. Hussain, H. El Enizi, A. Al Sagheer, D. Mohammad, M. Al Hajri, B. Al-Otaibi, and A. Al-shamali, Kuwait Oil Company; I. A. Likhoded, M. Harith, S. Sabat, N. D. Vasconez, S. Biniwale, S. Khataniar, M. Ahmed Elfeel, X. Garcia-Teijeiro, R. Xu, and S. Kumar, SLB

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229785-MS](https://doi.org/10.2118/229785-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper explores the application of AI to enhance the efficiency of waterflooding operations. AI can be game changing, but it relies on well-designed applications that consider many factors, not necessarily technical, to generate and sustain maximum value. The paper documents the approach and outcomes experienced during the piloting of an AI based solution that will benefit others who desire to make this transformation to AI enabled waterflood management. In particular, the paper highlights how AI applications have been integrated into Kuwait Oil Company's (KOC) existing waterflooding management procedures to drive operational and economic efficiency in the Wara reservoir of the Greater Burgan oilfield.

Introduction

Discovered in 1938 and produced since 1946 the Wara reservoir in the Greater Burgan (GB) Oilfield is currently being actively developed through infill drilling and waterflooding. The GB field is the largest clastic reservoir in the world with multiple stacked reservoirs consisting of mainly the Burgan and Wara sands. Among them the Wara is the most heterogeneous. Both the Burgan and Wara share a common water-oil contact but unlike the Burgan sands the Wara reservoir experiences limited support from the aquifer. The Wara reservoir pressure declined and quickly fell below the bubble point. Gas injection was initiated in 1962 but continued decline in pressure necessitated curtailment of production in 2006 prior to increasing offtake with large-scale injection of water since 2016 [Al-Naqi, 2009]. Active infill drilling is currently underway to produce the reservoir under a pattern waterflood. However, early water breakthrough due to channeling has introduced water handling challenges and raised concerns about sweep efficiency and recovery.

To improve Wara waterflood performance KOC is seeking to leverage digital and AI technology. AI is expected to offer numerous benefits in driving efficiency, productivity and decision-making. However, transforming the potential of AI to real value is not straightforward. It is a journey that needs innovative thinking and a process of experimentation and evaluation. A feasibility study to leverage AI for tactical and operational decision-making was undertaken. In an operational environment, time horizons are short and rapid insights generation, quick unbiased analysis and improved short-term forecasting are needed for

good decision-making. Such an environment can strain human resources, especially when dealing with large fields, leading to errors and missed opportunities. Hence, automation though AI can play a major role.

A design thinking, human-centric approach within an agile innovation framework that supports rapid prototyping, deployment was adopted for solutions development [Khataniar, 2023]. Deployable prototypes and a proof-of-concept for AI applications have been achieved that have shown over 95% reduction in data pre-processing and validation through successful outcomes in the field as a result of the analysis and insights developed using the new tool.

This paper is divided into three parts. In the first part, we discuss the challenges faced by the Wara waterflood management team, in part two we present the development of a collaborative solution and in the final part we discuss the results from its application before arriving at conclusions.

Wara Waterflood Management Challenges

The GB field is divided into fourteen sectors for management purposes. These sectors roughly align with reservoir blocks formed by faulting. A group of reservoir surveillance engineers are responsible for the daily monitoring of all sectors. They are supported by a field development group that provides subsurface related insights. Currently, 9 sectors covering the flanks of the dome type structure are under waterflood with varying degrees of maturity. Sectors 4 & 5 were selected to validate the proof-of-concept solution as they are representative of the challenges faced in Wara [Figure 1].

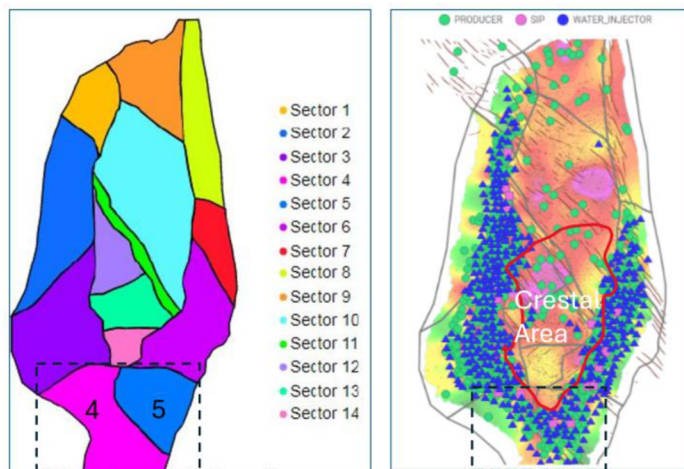


Figure 1—(Left) Map highlighting management sectors for pilot and (right) overview of water injection and production wells in WARA reservoir overlaid on a map of prevailing static pressure showing pockets of depletion and location of secondary gas cap.

The strategic plan for the development of the Wara reservoir has identified several reservoir management and operational challenges and strategies to address localized pressure depletion, formation of secondary gas cap, non-uniform water encroachment and bypassed oil [Al-Samhan, 2024]. Enhancing production from the crestal part of the Wara reservoir encompassing the gas cap while repressurizing to shrink the gas cap as much as possible is a recommended strategy. This calls for a significant investment in drilling and boosting facilities to handle water injection, production, treatment and disposal.

Heterogeneity

The Wara reservoir is characterized by multiple stratigraphic zones arising from its fluvio-deltaic origins and observed dynamic behavior. There is evidence of inter-reservoir communication from Burgan to Wara reservoirs through fault zones. A zonation study conducted in sector 3 proposed interpreting a detailed geological layering scheme for the Wara reservoir [Bowman, 2017].

The complex sequence stratigraphic layering scheme has been rationalized into two reservoir management units (RMUs): a Lower Wara (LW) RMU and a Middle Wara (MW) RMU. In some parts of the reservoir the Middle Wara RMU can be further sub-divided into an Upper Wara (UW) RMU. These units exhibit limited vertical and lateral communication due to internal baffles and barriers.

The complex reservoir architecture creates tortuous flow paths and pressure support from injectors may not reach distant producers effectively. Figure 2 shows a cross-section of wells around an injector centered waterflood pattern and the level of heterogeneity. Juxtaposition of different channel fills creates small-scale compartments that can hold significant pressure differentials (>100 psi).

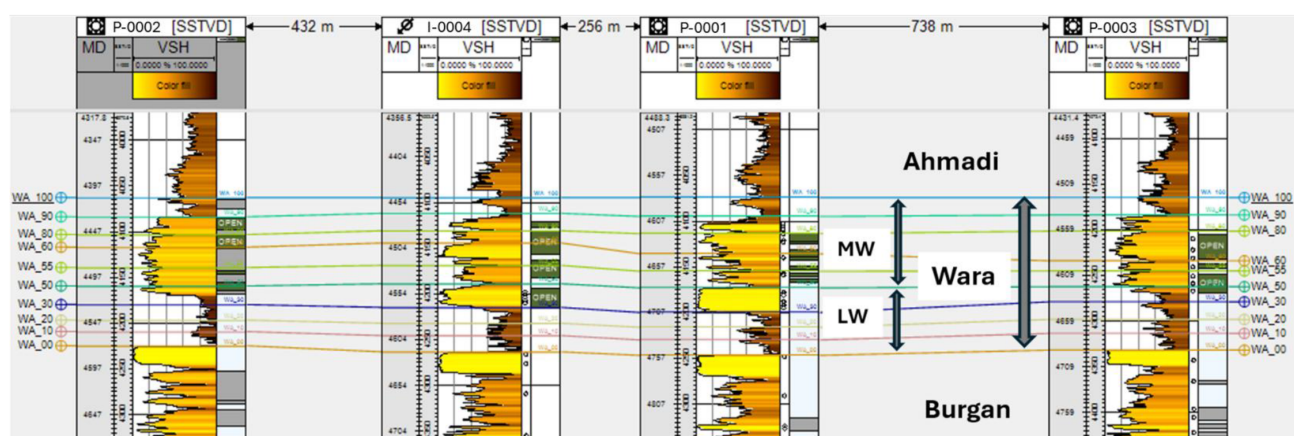


Figure 2—Well section showing major GB reservoirs, zonation, RMUs and lithology variation in a waterflooded well pattern.

This leads to uneven pressure depletion and crossflow between RMUs, complicating sweep patterns. The implications for water flooding strategy are the need for selective perforation to avoid ineffective injection and rapid pressure decline, injector-producer conformance tailored to the connectivity and quality of flow units, and artificial lift and GOR optimization due to uneven pressure support. A tracer study confirms the level of inter-well heterogeneity [Al-Ghanim, 2018].

Heterogeneity also complicates the ability to achieve good injection conformance and reduce water recycling [Al-Murayri et al., 2022]. Overall, the heterogeneous geology of the Wara reservoir presents significant challenges to waterflood optimization.

Operations

Currently there are over 1500 boreholes that penetrate the Wara reservoir and around 600 active producers and more than 400 injectors. New wells are introduced every month and recompletion activity is high as is expected in a field with many stacked reservoirs and zones. The Wara waterflood is being managed through monitoring of injector VRR. As water production increases, the need to reduce water recycling becomes as important as maintaining productivity through pressure maintenance.

In Wara, injectors and producers are planned to be drilled in a 5-spot pattern. Generally, wells belonging to a pattern are drilled from a single pad in sequence. Drilling operations are closely monitored to optimize pattern development [Al-Resheedi et al., 2021]. Some new wells intended to be producers were found to be swept (PNC logs) by earlier generation peripheral injectors either due to delays in drilling the wells since planned or due to unexpected heterogeneity or both. Planned producers have been converted to injectors requiring an overhaul of the entire pattern injection and offtake recommendations.

Surveillance Engineering Tools

Several pilots have been undertaken to digitalize and provide production support systems for the GB field by the SEK Asset [Dashti et al., 2022; Desai et al., 2019; Rane et al., 2018]. These systems have mainly

targeted production wells and surface facilities. Waterflood surveillance and optimization needs to focus on the connectivity between injectors and producers or waterflood patterns.

The current tool used by the Wara waterflood management team is very simple and relies on subjective input from the user. Production data availability is known to lag, sometime up to 3 months. The recommendations are difficult to verify as there is no ability to generate forecasts and run optimization scenarios or what-ifs. The process is slow, tedious and manual greatly risking poor waterflood performance and missed optimization opportunities. Further discussion on the tool and workflow is provided below.

Collaborative Solution Development

The design thinking human-centric and iterative approach to solution development is underpinned by collaboration among all stakeholders. Design thinking places the user at the heart of the process, focusing on their needs, behaviors, and pain points through direct observation and engagement. We use the innovation process framework incorporating design thinking [Khataniar et al., 2023]. The process involves a circular sequence of steps: Discovery, Define, Ideate, Experiment, Build & Test, and Implement.

Discovery

Users were interviewed, and their daily routines and software used were analyzed to understand and prioritize the most important pain points. The software tool supporting the reservoir engineer (RE) is an application with dashboards hosted on a commercial information management system (Visualization Dashboards). Inputs to the application such as production/injection data are handled manually. The RE is dependent on other subsurface and production domain experts to complete the inputs.

The main waterflood performance KPI monitored is voidage replacement ratio (VRR). Injector VRR is computed by selecting producers within a limiting radius from injectors. The production allocation fractions that allocate a wells production back to the injector are provided manually by the user. Injectors that are operating at a low VRR are analyzed for opportunities to increase injection rates to achieve target VRRs. Similarly, injectors with VRR greater than the acceptable range are recommended for rate reduction to operate in the desired range. Recommendation memos are then issued to production operation teams for field implementation.

Analysis at the individual Wara RMU (MW and LW) level is conducted by using split factors to back allocate well production. The allocation and split factors provided as input by the RE are stored in an MS Excel worksheet on a shared drive accessible by the field development team members. Split factors are manually assigned based on the most recent ILT/PLT survey, if available, otherwise the engineer provide their best estimates based on consultations with supporting team members. Allocation factors are also set arbitrarily in the spreadsheet after consultations with the geologist who qualitatively assesses potential connectivity levels between neighboring injectors and producers.

At KOC, a commercial seismic interpretation, reservoir modeling and simulation platform is used by the SEK Assets Field Development team. Commercial reservoir simulation tool is used for strategic planning. 3D geological and reservoir simulation models are constructed and history matched to evaluate development strategies and Asset Action Planning (AAP) activities [Al-Samhan, 2024; Filak, 2017]. Full-field model project.

However, the full-field model is not suitable for use in waterflood monitoring operations. Field, sector and gathering center level history match requirement is prioritized over well level to align with the objectives of AAP. The quality of history matching at the well level is not sufficient to condition the geology for reliable injector producer relationship analysis. The size of the model, computing requirements and frequent updates to the model due to an active drilling and completion schedule makes it virtually impossible to use the full-field model for waterflood monitoring.

A separate project is maintained for reservoir management purposes on the commercial modeling platform. This project is not used for 3D modeling but instead used by members of the field development team responsible for reservoir surveillance and management. The project is regularly updated with new well and surveillance data. This data is used to update surveillance maps (net thickness, water encroachment/flood front and pressure) at the RMU level on an annual basis.

Defining the "To-be" State and Persona Identification

Following discovery, the problems encountered by the RE were systematically defined in terms of needs and concerns. Ideation sessions were held to generate possible solutions that would make users happy. The essence of a "To-be" state is a reliable decision support system that automates the data processing steps, provides efficient visualization for rapid analysis of large datasets and hybrid AI-Physics models to gather fast, reliable insights and perform short-term forecasts for comparing decision alternatives. The premise for models is to help produce a rolling quarterly forecast every month.

Possible solutions identified incorporated criteria such as easy-to-use, easy-to-deploy and sustain, reliable insights and recommendations, responsive software and maximum automation. KOC mandated a few requirements: physics-based approach to complement AI models, solutions to be deployed on-premises, visualization dashboards to continue to use the existing commercial platform and follow cyber security policy and prohibit the execution of computational scripts on local user machines.

The generated solution ideas were feasibility checked through experimental prototyping and either updated, discarded or modified based on immediate feedback from users. Following several iterations of discovery-define-ideate-experiment a "to-be" state that satisfied all stakeholders was achieved. The desired state involves the use of various models (hybrid AI, physics, analytical, heuristic) to estimate split factors and allocation factors, generate production forecasts and run optimization scenarios.

AI-Waterflood Modeler. Analysis of existing waterflood management work practices identified several personae with distinct roles, responsibilities, needs and concerns. The introduction of a model-based (AI / Physics) support solution required a new persona - the "AI-Waterflood Modeler" (AWM). The role of this persona requires data science skills not available in the current team. The AWM persona would be responsible for building, maintaining data-driven hybrid AI-Physics models, supporting the RE. This person would be knowledgeable in reservoir engineering, have a good understanding of the commercial AI platform to be deployed and python programming skills. Figure 3 shows the key persona, their related roles and responsibility.

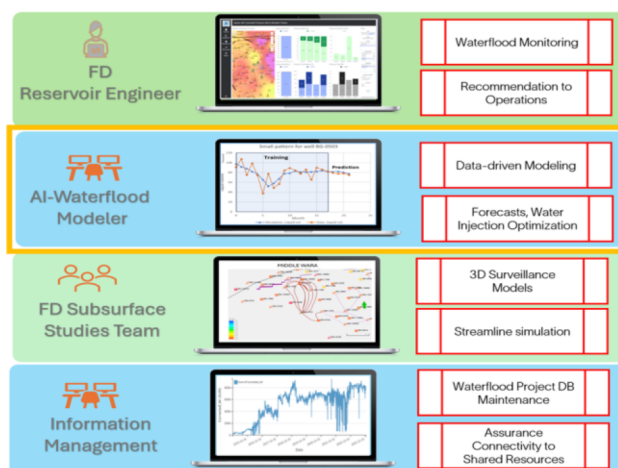


Figure 3—Persona roles and responsibility including the new role of the AI-Waterflood modeler.

Solution Architecture and Components

The solution architecture illustrated below in Figure 4 is structured into three major functional layers (Khataniar et al., 2023):

- Advisory
- Model
- Data

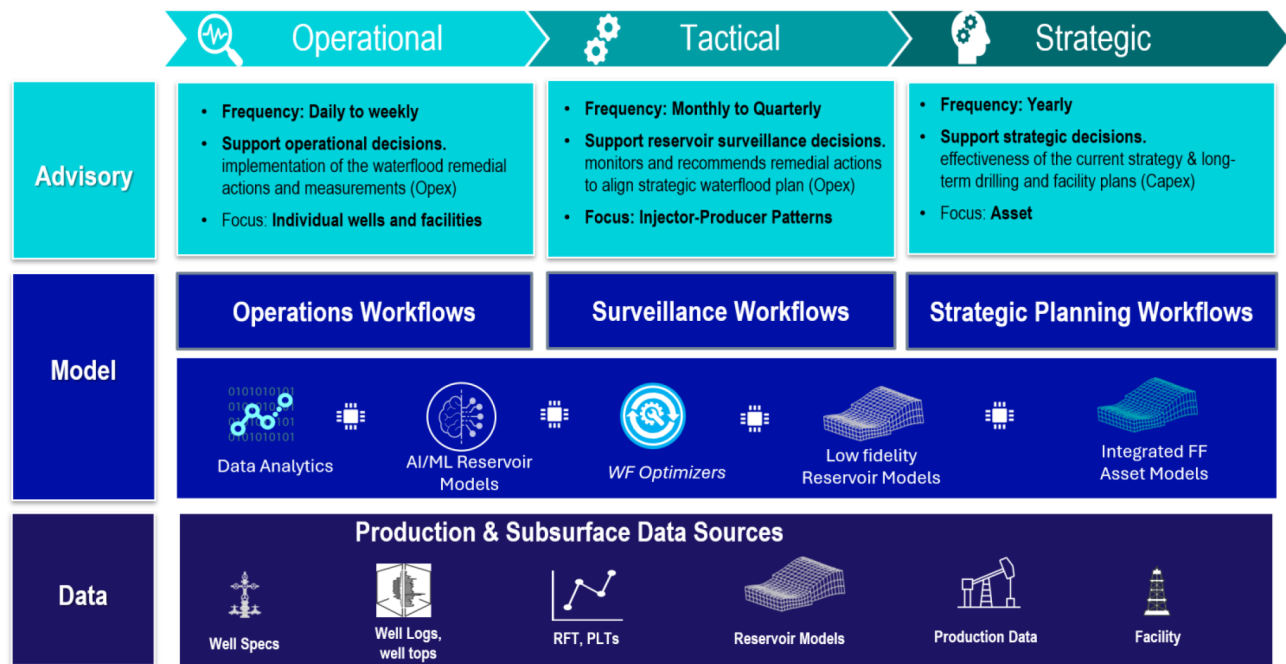


Figure 4—Solution architecture for waterflood management decision support system showing three categories of decision situations and three tier layered system.

The layers broadly map to the different persona identified earlier. This architecture is easily extensible to cover future requirements. We include a brief of each layer, and the components prototyped.

Advisory Layer. The advisory layer and its components are designed for the RE. The RE will be able to monitor general waterflood performance daily (subject to daily data processing), identify sub-optimal performing wells and patterns and inform the AWM. The RE will then review the recommended actions proposed by the AWM and may accept them or suggest evaluating other what-if scenarios. The user experience and interface design were done in close consultation with a representative of the SEK asset REs.

Multiple dashboards and QC reports have been implemented based on project requirements, providing near relevant-time insights into the data. All required data tables are dynamically linked to a SEK WF Project Database (Oracle) in the back end. The dashboard is refreshed daily, and the data cache is available to all REs assigned to the different sectors.

An overview of field, sector, well and zone performance with various filters allows tailored views. Displays are interactive and include maps, line plots, pie-charts and histograms. Injector-centric and a producer-centric visualization dashboards with a map display of the selected injector or producer and connected wells are used to display split factors and injection, production rates and allocation factors by zone. The injector-centric dashboard displays injection allocation factors and water production information to give a qualitative view of water recycling. The producer-centric dashboard provides a similar display but shows production allocation factors and VRR by injector. A dashboard to visualize recommendations made

by the optimization routine has been created. Injectors with low injection utilization or water recycling will be flagged and rate reduction recommendations made.

Separate dashboards to quality check data used in analysis and recommendations have been provided, e.g. QC historical split factors with colour coding to identify the method applied (permeability-thickness or PLT/ILT), perforation and production/injection data alignment, wells with missing tops, permeability, etc.

Model Layer. The Model layer was defined based on the needs of the AWM. The model layer consists of WebApps, python notebooks and temporary data storage hosted in the commercial AI-integration platform. The key features of the model layer include a) automated generation of split factors and zonal allocated production rates, b) a hybrid AI-Physics Data-driven Modeling application that generate allocation factors and production forecasts and c) an injection rate optimization algorithm. In addition, algorithms were coded to generate allocation factors using variants and combinations of geometric, analytical and heuristic methods. In the following sections the features are briefly described.

Split Factor Estimation

ILT and PLT surveys provide the best information to estimate split factors [Woodling, 1993]. However, not all wells are surveyed, and the low frequency of surveys means that data may not be current. Based on steady state inflow performance theory, permeability-thickness is often used in an indirect approach to computation of split factors.

The permeability method depends on the availability of reliable petrophysical data, well tops, permeability, net sand, perforation length, etc. Many wells lacked well tops; perforation depths and completion IDs were inconsistent or completely missing and permeability logs were missing in some wells. New wells and completions always pose a challenge due to time required for petrophysical and geological analysis.

Various technologies including AI have been leveraged for interpolating well tops, performing perforation data quality checks, permeability log imputation and appropriate averaging of permeability over relevant reservoir thickness.

The workflow was tested on historical data and automated via live connections to the KOC database. The chart below shows the split factor workflow [Figure 5].

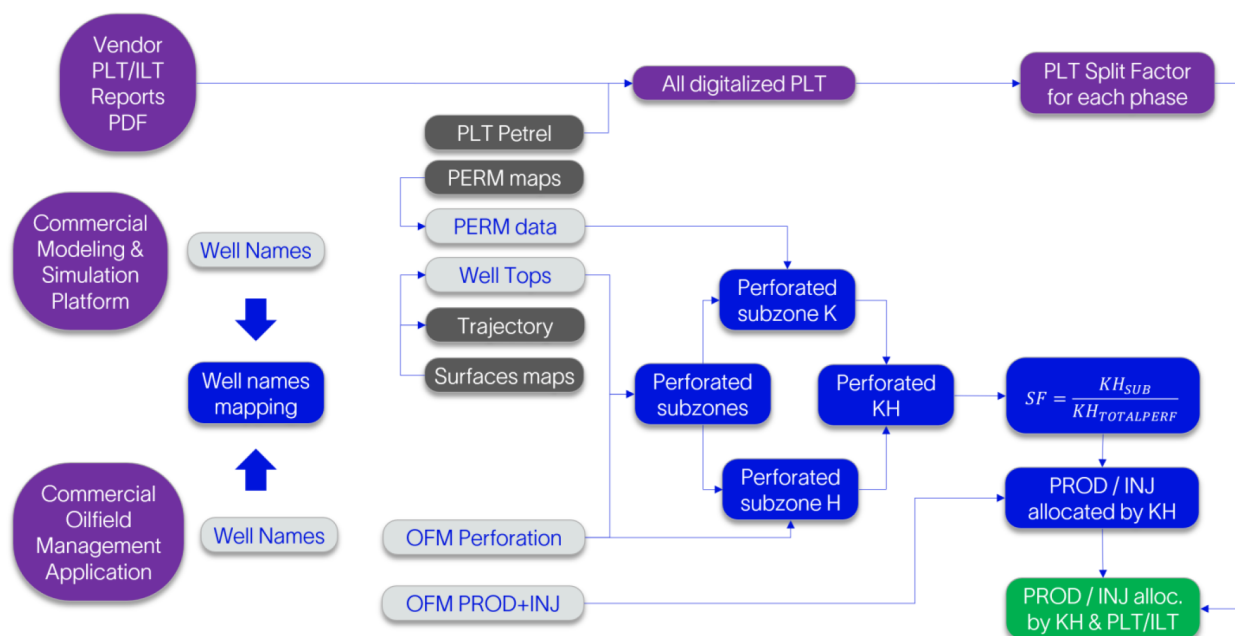


Figure 5—Split factor estimation workflow.

Allocation Factor Estimation Models

Understanding of injector-producer relationships lies at the heart of any waterflood optimization solution [Woodley, 1993; Grinestaff, 1999]. Due to the dynamic nature of the reservoir, changing pressure distributions and ongoing drilling/completion operations, the relationship is never static and changes with time. Reservoir heterogeneity usually has a first order effect on the dynamic connectivity between injectors and producers.

Production allocation factors are fractions of the total well production assigned to each injector-producer connection. Injection allocation factors are fractions of the total injection rate assigned to each injector-producer connection.

Unlike split factors that can be measured directly using PLT/ILT survey there is no direct way to measure allocation factors. They are usually estimated using various methods. Table 1 presents a high-level comparison of such methods against criteria that are required for tactical and operational situations.

Table 1—Comparison of allocation factor estimation and production forecasting models and their suitability for use in operations environment.

Models	Category	Reliability of Afs	Data Needs	Speed	Automation Friendly	Power Required
Simulation (Fully Implicit, Sequential Fully Implicit)	Traditional Full-field Numerical Models	Depends on accuracy of underlying physics model.	Highly complex dataset, requires expertise to build, history match and update	Slow (Hours)	Low. Difficult to keep updated for surveillance operations	Cloud, HPC required.
Surrogates of full-field simulation models (NN)	Surrogate (simulated data-driven) model			High (Minutes/Second)		Cloud, HPC required to generate simulated data for surrogate training.
Simplified physics (2D CV-RB, 3D streamline, Multiscale, etc)	Reduce-order model	Depends on representativeness of ROM		High (Minutes)		Cloud/HPC may still be required to generate simulated data for training surrogate. Local desktop possible.
Surrogate of ROM (NN)	Surrogate (simulated data-driven) model			High (Minutes / seconds)		
Analytical	Simple Algebraic equation	Assumptions maybe restrictive. Depends on data quality. No forecasting capability	Geological maps (fluid, well rates optional)	Very High (seconds)	High.	Low power desktop/laptop. Minimum computing requirements.
Hybrid, Physics (CR) and Multivariate linear regression, PINN.	Reduced-order (Measured data-driven) model.	Depends on data quality.	Simple dataset. Requires production/injection rates, BHP only.	High (Minutes / seconds)	Medium. Easy to update but may require re-training.	Workstation. Depends on number of datapoints and features.
Multivariate Linear Regression	Measured Data-driven model	Fixed pattern assumption maybe restrictive. BHP not used.	Simple dataset. Requires production/injection rates only.		High.	

The categorization and comparison of modeling techniques is a complex subject and the table by no means represents a generalized analysis. In many areas, such as reduced-order models (ROMs), and the application of advanced AI, active research is ongoing [Nagao et al., 2024, Yu et al., 2021]. The terms data-driven, data-physics, etc. are often used to mean models that are derived by fitting either real measured or reservoir simulation model generated data. The reservoir simulation model may in turn be calibrated/history matched to the measured data. Figure 6 shows a schematic to distinguish the two ways that AI models could be constructed.

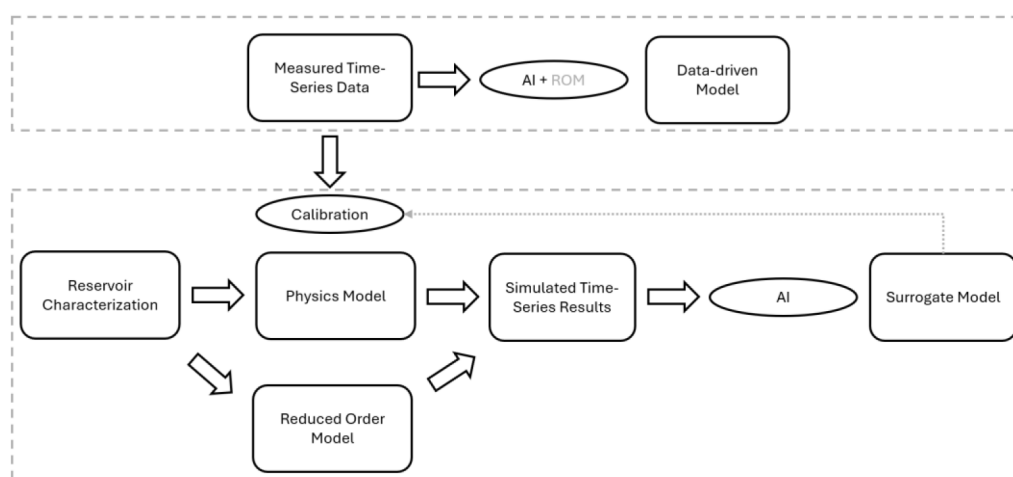


Figure 6—Schematic comparison of processes for constructions of DDMs from a) observed data and b) calibrated simulation results

"Measured" Data-driven Models (DDM).

The difficulty of availing a well-level history-matched reservoir simulation model and keeping it up to date ruled out the feasibility of building AI-surrogates. A DDM has been deployed earlier to manage a waterflood in Ecuador [Khataniar et al., 2023, Grijalva et al., 2023]. This DDM is based on simplified "capacitance-resistance" physics [Ali-Yousouf et al., 2006; Sayarpour et al., 2006, Holanda et al., 2018] and a multi-variate linear regression modeling approach. However, the DDM required further upgrading to address the larger well count and multizone RMUs in the Wara reservoir.

The interaction between a single injector-producer well pair is constrained to honor material balance and fluid flow laws. Production and injection rate data are treated as time-series signals and the interwell connectivity is determined using multi-variate linear regression. The method does not need geological information to infer what the waterflood pattern. The reservoir and fluids in the inter-well space between the well pair affect the arrival time and strength of the injection signal at the producer. A hydraulic discontinuity or high system compressibility may mean that an injector signal never arrives at the producer or is seriously attenuated to avoid detection. In waterfloods, however, the system compressibility is usually small. The inter-well connectivity strengths are processed to generate production and injection allocation factors [Elfeel et al., 2024]. Injection loss and other sources of production such as changes in bottom-hole pressure, depletion or aquifer influx can be estimated. A WebApp component was developed to support the modeling of DDMs.

Analytical models

Depending on the quality and sufficiency of the production data it may not always be possible to apply DDM always. Especially, new wells have no history to fit. Analytical methods have existed since waterflood were subjected to engineering analysis [Saadatpour, 2012, McElhaney, 2016].

A new analytical algorithm has been developed together with KOC. The method proportions allocation factors to I-P pairs based on the inverse of distance squared. Further weighting can be done using average I-P pair transmissibility and flow rates [Figure 7]. The method is extremely fast to compute; however, it has some restrictive assumptions and limitations. The method remains to be verified, and results should be validated by other methods.

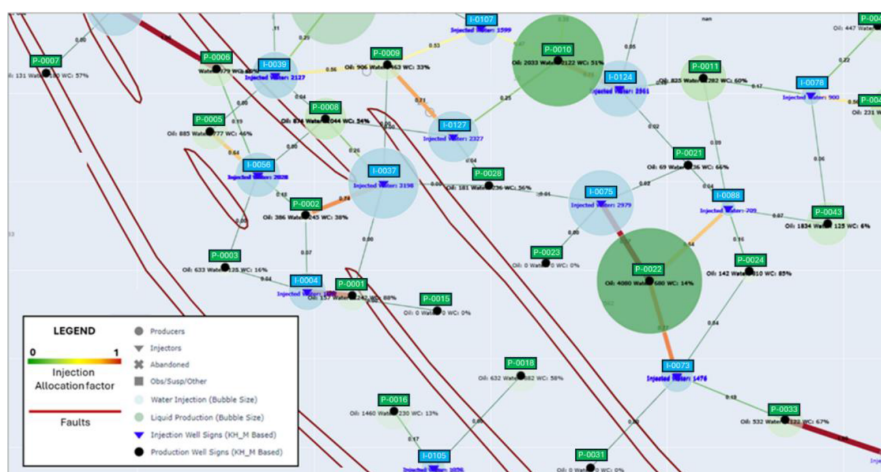


Figure 7—Analytical injection allocation factor estimates from petrophysical properties (connectivity between injector-producer pairs).

Streamline ROM

Tracing streamlines or simulating flow in streamlines in a reservoir model allow the generation of allocation factors [Grinestaff, 1999; Woodling et al., 1993]. Total allocation factors can be generated easily and quickly with representative single phase fluid models. An experimental prototyping exercise was thus undertaken to investigate the feasibility of exercising this option.

A streamline ROM based on a 3D layer cake representation of the Wara reservoir for sectors 4 & 5 was constructed. A detailed 10-layer stratigraphic zonation, which provides much more detail than the RMUs, and better represents smaller sand bodies was captured in the geological model. Further grid refinement of up to 100 layers was used to capture precise location of perforations. The 1.5 million cell model computed allocation factors as at the current day in about 45 seconds on a simple desktop computer. The workflow is partly automated; it is easy to update the model with the most recent well information, production/injection and surveillance maps [Figure 8].

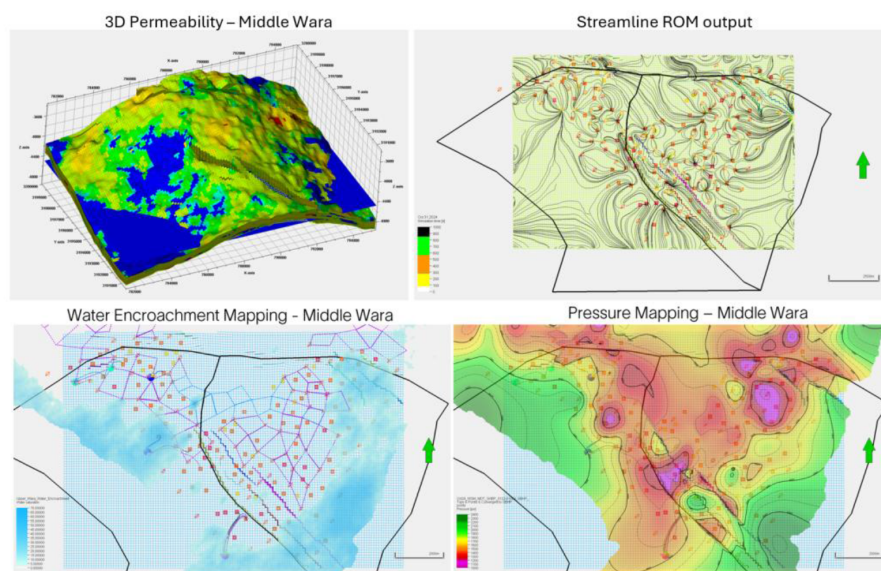


Figure 8—Streamline reduced-order-model for MW RMU sectors 4 and 5 with input permeability, water saturation and reservoir pressure maps.

Assessment of the streamline ROM was done to study the effects of reducing the flow physics and boundary conditions. The analysis shows that the single-phase model compares well with the two-phase model both in terms of the pattern configurations and estimation of allocation factors [Figure 9]. Finally, streamline ROM can be executed with well bottom-hole-pressure controls to estimate zonal production rates or split factors.

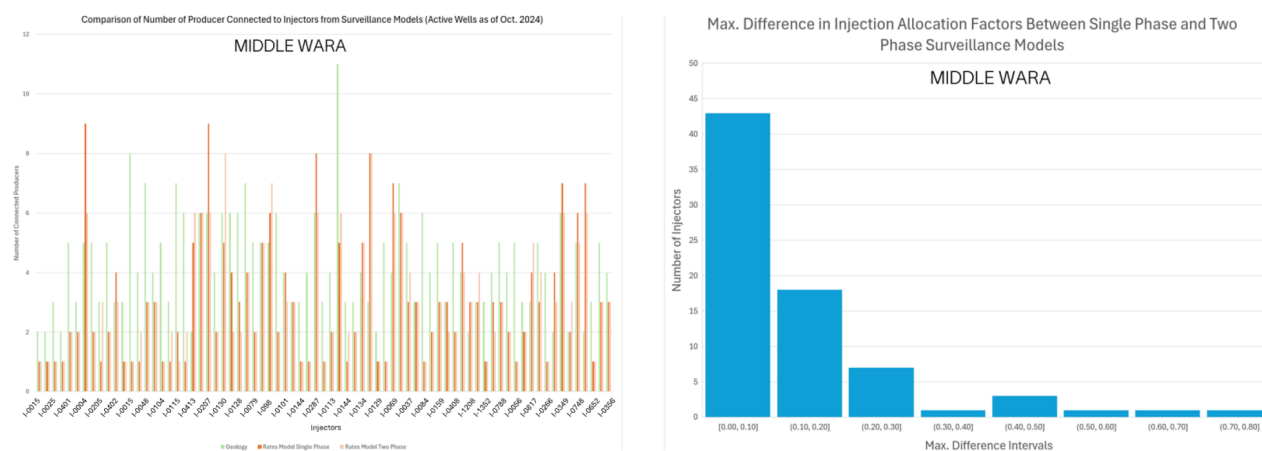


Figure 9—Sensitivity of streamline ROM results with gradual reduction in physics. (Left) variation in connected producers (injector centered pattern) and (right) difference in allocation factors.

Water Injection Optimization

In a waterflood it may be necessary to produce at high water-cut levels to maximize recovery as long as the economics of the operation permit. However, in heterogenous layered reservoirs like Wara, it is possible that some layers are completely flooded resulting in injected water recycling through producers and not contributing to recovering additional oil. Recycling of water could also happen because of bypassing recoverable oil within a layer.

The "utilization of injected" water KPI is often used to monitor injector performance. It tracks the amount of oil produced by a well per unit volume of injection supporting its production. The pattern flood management (PFM) feature is a reservoir engineering-based routine implemented in commercial reservoir simulators [SLB, 2022; Spyrou, 2017] that offers waterflood optimization strategies based on KPIs. PFM logic has been implemented in a WebApp and adapted for use with DDMs.

Specifically, the Koval method was used with the DDM to fit the entire range of water cut data from breakthrough to flood out using a physics based fractional flow model [Cao et al., 2015].

Data Layer

The data layer consolidates and preprocesses a wide range of inputs such as wellbore specifications / completions, production and injection data, PLT / ILT logs, RFT data, well tops, and various geological data from static reservoir models (Figure 10). This data is automatically ingested and quality-checked via high-performance pipelines, with live links to operational databases (e.g., Sahala) and AI-powered OCR routines used to digitize image-based PLT and ILT data.

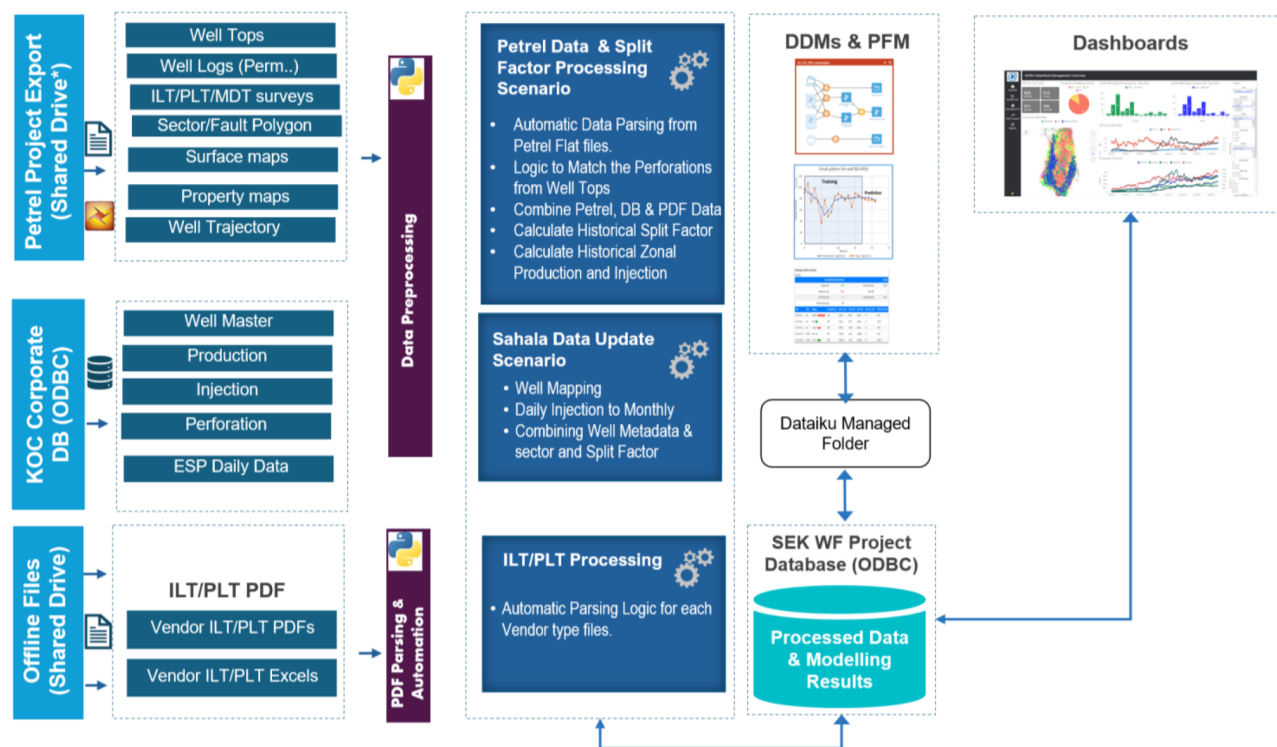


Figure 10—Schematic of data layer showing data sources, processing and automation scenarios, and consumption by models and dashboards.

To ensure seamless data updates and workflow execution, the solution leverages AI-platform scenarios for automated data ingestion, processing, and transformation. Two key time-based AI-platform scenarios have been created to manage the daily updates from Sahala and Petrel data sources, ensuring accurate and up to date information is available on visualization dashboards.

The Sahala Data Update Scenario is responsible for updating all Sahala database records including production, injection, well master, and perforation data is scheduled to execute at 22:00 p.m. daily.

The Petrel Data and Split Factor Processing Scenario is scheduled for 02:00 a.m. daily. This scenario automates the processing of all Petrel geological data exported as flat files and executes split factor workflow processing, handling allocation and zonal split factor calculations, and updates the SEK WF Project Database with the processed results.

Results

Automation

The data layer of AI-integration platform scenarios was able to preprocess data for visualization in dashboards and prepare time-series for all models on average in about 2 hours. The Sahala Data Update Scenario takes about 25 minutes to complete, and the Petrel Data and Split Factor Processing Scenario takes about 1.75 hours. KOCs user recall that on average this process would have taken 4 to 5 days or even more with the legacy system. Hence, by this automation an efficiency gain of over 95% has been achieved. The structured database has made a remarkable change in responsiveness compared to the legacy system.

Multi-zone waterflood management requires reliable estimation of split factors. A refined permeability-thickness ratio method produced results consistent with PLT/ILT analysis. AI methods are used to generate missing petrophysical data. In about 75% of the wells the difference between surveys and the permeability method was less than 20%. In 90% of the wells with PLT surveys the difference between the total fluid and water split factors was less than 20%.

Decision Support

The introduction of fast AI augmented models for waterflood surveillance and analysis has added a new dimension to the field reservoir engineer. These models can detect waterflood patterns based on geological and time-series well information. In the legacy system, the operations geologist provided geological interpretation to estimate pattern configurations and allocation factors. In the new system, this is objectively done using map-based models, streamline ROMs and complemented by pure measure data-driven modeling methods. This provides a very balanced view and insights into the pattern behavior removing bias and addressing uncertainty.

A new analytical method has been introduced to estimate allocation factors based on geological input adding a level of redundancy in case the other methods are not suitable. The method is intuitive and would be easy to use by most field reservoir engineers pressed for time. However, it has some limiting assumptions and heuristics that need to be considered when using the results.

As a stress test, DDMs were trained for all well patterns in sectors 4 & 5 in the order of 5 to 10 minutes depending on the parameter and optimization settings. There is a need for human intervention and iterations to refine the model fitting hence this does not represent the true time to prepare a good set of models. Nevertheless, the test provides information that proves that DDM performance lies within the range required for operational purposes.

The streamline ROM model designed for operational purposes was tested on sectors 4 and 5. At the time, these sectors represented 16% of the active flowing well stock in the Wara reservoir. The two-phase model computed allocation factors in less than 45 seconds. The single-phase model required even less time. Comparison of allocation factors with DDMs revealed interesting lines of investigation when values did not match.

The adaptation of the PFM optimization logic with DDMs has been successfully tested with known reservoir simulation problems and results.

Field validation

A prototype implementation of the custom solution was piloted in Sectors 4 & 5. Insights generated from the solution were used to support a decision to carry out an intervention in a producer (P-0001). A multi-model analysis combining DDM, streamline ROM, and analytical method consistently supported a preliminary surveillance diagnosis that most of the injected water from a nearby injector, I-0004, was being short-cycled to P-0001 in the MW RMU. Intuitively,

Figure 11 shows taken from a dashboard display of surveillance information for the injector-centered pattern around I-0004. This is based on a fast, quick-look analysis of the waterflood patterns using the analytical method estimated allocation factors. It reveals a strong connection between I-0004 and P-0001, as would be expected given the proximity of the wells.

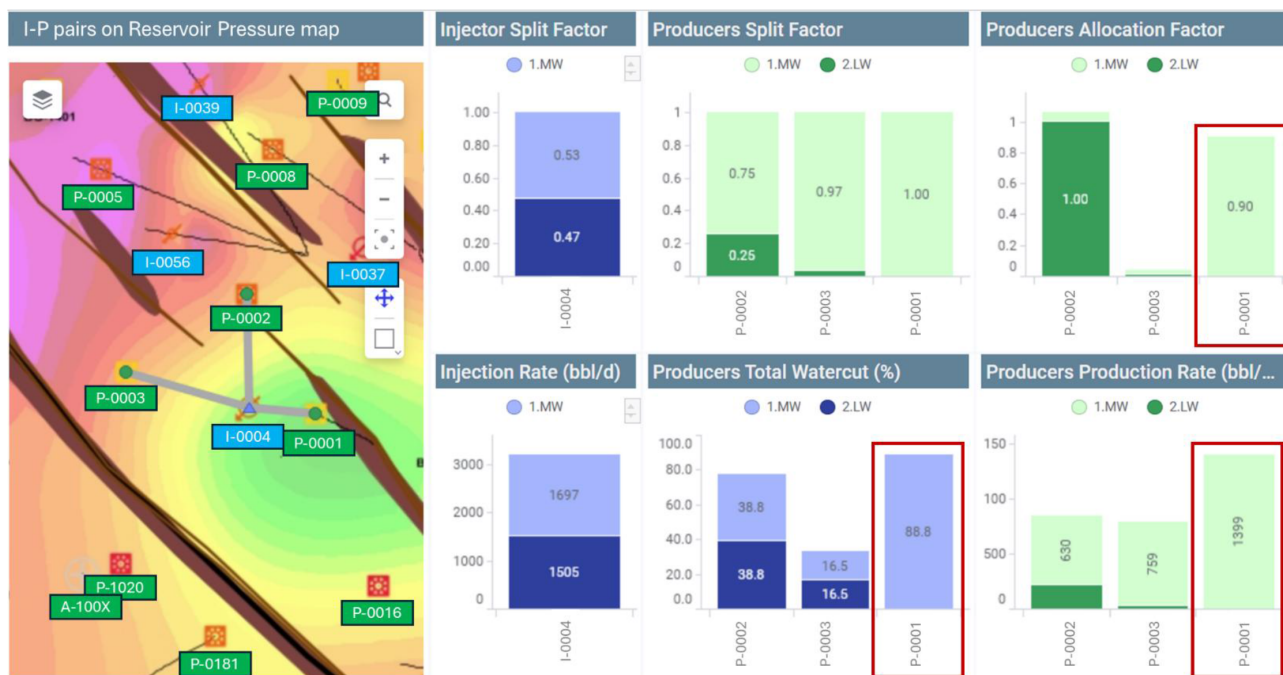


Figure 11—I-0004 water injection pattern surveillance showing RMU MW and LW performance (from visualization Dashboard).

Figure 12 shows the streamline ROM display of interwell connectivity before well intervention. Split factor estimates suggest the injector I-0004 is injecting equal amounts of water into both the RMUs. However, the patterns are quite different. In the MW, the connected producers are P-0001 and P-0016, and in LW, P-0002 shows good connectivity while the connection to P-0003 is weak. The high water cut (~90%) in P-0001 and good pressure support (pressure distribution maps underlying the streamlines display) are indicators of water recycling.

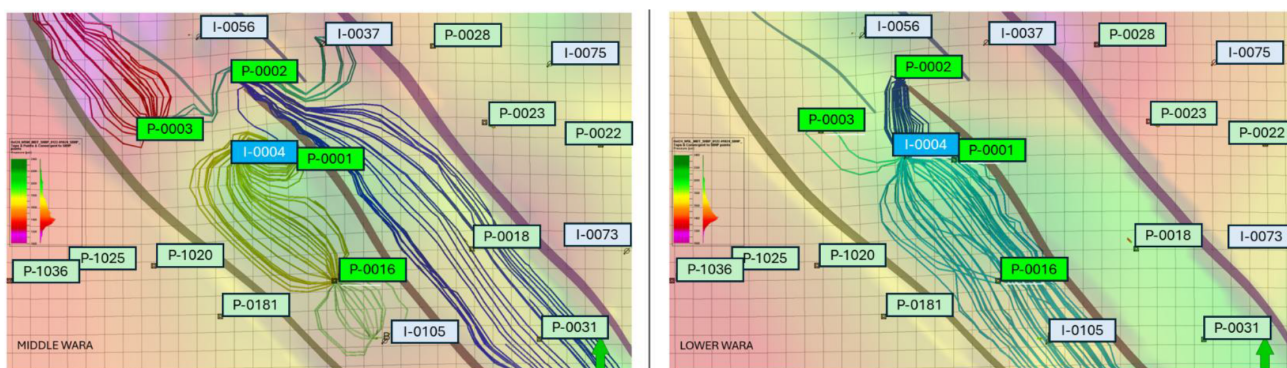


Figure 12—Streamline ROM sector 4 & 5 showing I-0004 water injection pattern (left) Middle Wara and (right) Lower Wara.

DDMs of producers P-0001, P-0003, P-0002 and P-0016 were built. A 1-year training period from January 2024 to February 2025 was used. The results of model fitting for P-0001 are shown in Figure 13.

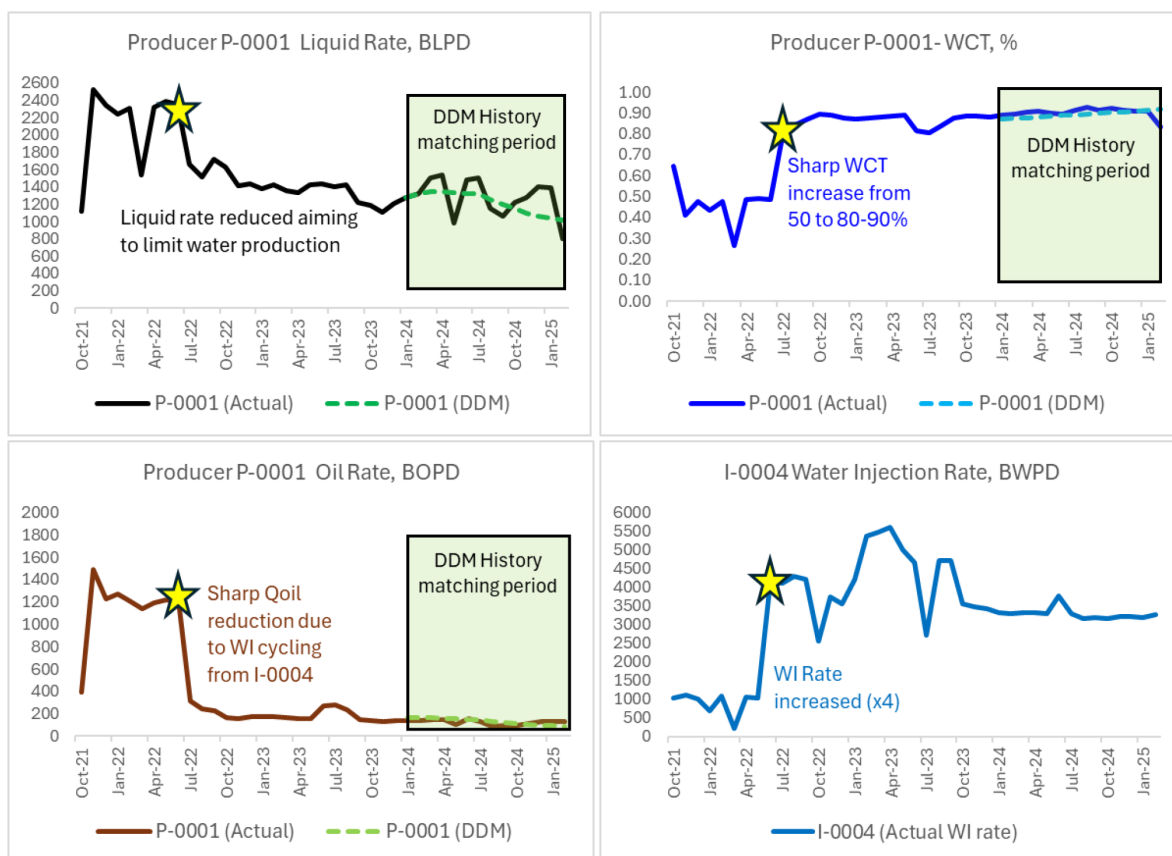


Figure 13—P-0001 oil and water production rates history and WCT with DDM history matching results (dotted lines) and I-0004 historical water injection rate.

The historical analysis of Producer P-0001 clearly reveals a case of direct water cycling from Injector I-0004, confirmed independently through all three methodologies: Analytical Allocation Factors, the Data-Driven Model (DDM), and the Surveillance streamline-based model. The issue began around mid-2022, when the injection rate from I-0004 increased from ~1,000 to 4,500 BWPD. This coincided with a sharp and sustained water cut increase in P-0001 from approximately 50% to over 90% as seen in the top-right plot. In parallel, the oil production rate of P-0001 dropped sharply, from over 1,500 BOPD to under 200 BOPD, indicating ineffective sweep and water breakthrough.

The liquid production rate from P-0001 was subsequently reduced to limit water handling, but the high WCT persisted, further validating the presence of a strong hydraulic connection. Notably, this behavior was captured with a high level of agreement by the DDM history-matched model, which used only monthly injection and production data from the most recent year.

Despite the coarseness of the input data, the model demonstrated reliable alignment with both liquid rate and WCT trends, underlining the robustness of the AI-based DDM even under sparse data conditions. The allocation factors, time and decline constants derived from the DDMs are cast from the perspective of the injector I-0004 in [table 2](#).

Table 2—DDM results for producers connected to injector I-0004.

RMU	Injector	Producer	Interwell distance	Injection Allocation Fraction	Time Constant (days)	Decline Coefficient
MW	I-0004	P-0002	492	0.221	1	50
MW	I-0004	P-0003	535	0.116	50	50
MW	I-0004	P-0001	250	0.367	1.4	50
MW	I-0004	P-0016	875	0.295	13.4	50

The allocation factors from the 3 different sources were reviewed and processed to an acceptable set to feed the pattern flood management optimization tool. Due to the difference in the estimates from different sources this step still requires some human judgement today. Notably, all 3 methodologies highlight domination of water injection from I-0004 to P-0001.

It also should be noted that P-0003 has the highest time constants with the lowest Allocation fraction, in fact this insight is supported with constant pump intake pressure decline in this well and GOR increase, indicating absence of pressure support for this well in current conditions. As a result, the PFM optimization engine recommended the following: reduce production from high WCT well P-0001 by 67% (almost stop the well), increase production in low WCT well P-0002 by 91% and reduce injection in I-0004 by 17% with an aim to achieve a VRR of 1.

The decision was taken to squeeze the MW RMU completion in P-0001 and recomple the well in the lower Burgan sand. At the same time maintain the injection level in I-0004 with the intent to provide more support for wells P-0002 and P-0003 in the MW RMU.

Following the implementation of the decision the I-0004 flood pattern demonstrated a significant improvement in production performance. As illustrated in Figure 14, the total oil production rate increased from about 800 BOPD to 850 BOPD in the period between February and May 2025. Most notably, the water cut was reduced from about 65% to 42%. The recompletion of P-0001 in Burgan reservoir yielded an additional 2,300 BOPD showing evidence that the use of the new solution supported good decision-making.

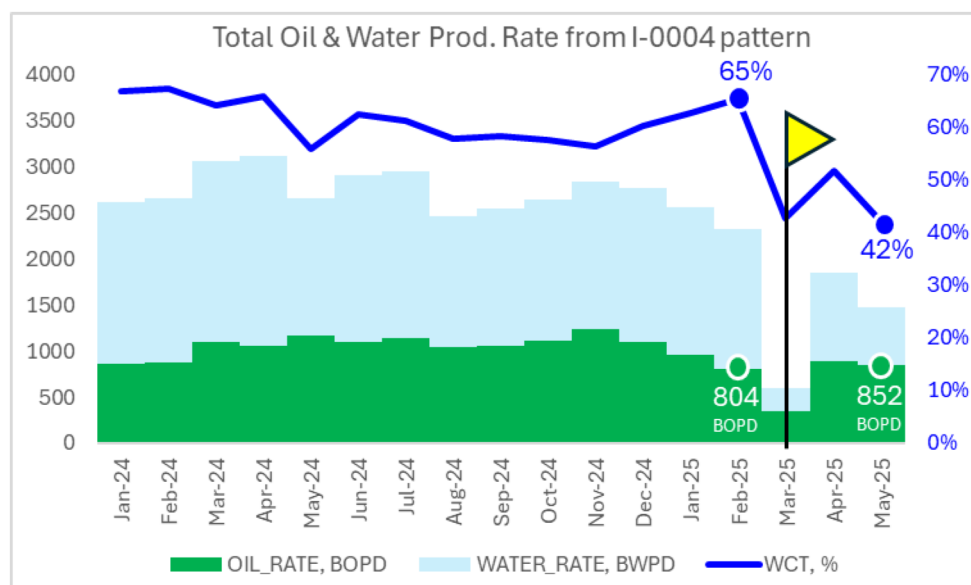


Figure 14—I-0004 injector pattern total oil and water production rates before and after recommendation implementation.

Discussion

The "measured data-driven" model implemented is sensitive to the quality of the observed data. At KOC currently, well production rate data is available as monthly average whereas injection rate data is available daily. Producer bottom-hole pressure availability is sporadic in natural flowing wells and frequent in well with ESPs. The injection rate data is averaged monthly to align with production allocations. The Wara pattern waterfloods started around 2016 and therefore the most mature patterns have about 108 data points at the maximum. Majority of the recent wells have far less data points. This affects the quality of the DDM fitting.

However, unlike pure data-driven AI models, the hybrid DDMs require much less data to get reasonable estimates of allocation factors. A thumb rule suggests at least four times more data points than model features/parameters. A factor that affects results is related to many patterns being in the fill-up stage of a waterflood. During this stage, the pattern pressure progresses towards the operating pressure ($\sim$ bubble point) and gas remaining in the reservoir leads to signal attenuation. The modeler should be aware of these issues and maintain oversight when training the hybrid AI models. As patterns complete fill-up the DDMs will produce better quality results.

Conclusions

The performance of a waterflood, like most assets, can be maximized through good decision making. In this paper, we focused on the tactical and operational decisions that affect the field reservoir engineer and improving the decision support system. AI is the next big technological revolution after the world wide web. KOC therefore commissioned a project to understand how best the power of AI and associated digital technologies could be harnessed for improved waterflood management. Sectors 4 and 5 of the Wara reservoir of the Greater Burgan oilfield were selected as the pilot area to execute a proof-of-concept project. The following are some of the findings and achievements of this project:

An on-premises prototype decision support software application was developed and tested on KOC's computing environment. Design thinking and agile development approach was adopted to build a customized system architecture that increased efficiency, productivity and decision-making. The highly iterative and collaborative solution development approach instilled user ownership in the solution and assurance for a successful deployment.

The early tool required a lot of manual pre-processing of data before it could be loaded into a dashboard application for analysis. Recommendations were made with simple analytics and arbitrary assumptions of waterflood patterns that did not capture the intense heterogeneity of the Wara reservoir. The new application leverages hybrid AI-Physics data-driven models, streamline surveillance models and improved data analytics. The use of geological model-based and time-series data-driven model-based estimation of flood patterns and allocation patterns offer multiple perspectives and deeper insights that serve to reduce bias and uncertainty.

A field proven streamline simulation-based optimization logic has been adapted for use with data-driven models and integrated into the decision support system. Short-term forecasting ability of models allows comparison of actionable alternatives before decision-making.

The new system has automated the pre-processing routine and reduced the time required to prepare for analysis by over 95%. In real terms that is a reduction in time days to hours. The jobs run daily at night and dashboards are updated at night ready for the reservoir engineer to monitor performance daily.

The system is being validated in sectors 4 and 5 and is already beginning to yield rewards. A recent water injection balancing recommendation to reduce recycling made using the new system has resulted in an increase in oil production by around 2300 bopd. The proof-of-concept project has shown some ways of how the power of AI can be harnessed resulting in an application being prototyped. The flexible architecture of the application and the AI platform used will serve to keep the application up to date with

the latest developments in AI technology such as the use of large-language-models (LLMs) for agentic AI applications.

Acknowledgement

The authors would like to acknowledge the patronage of KOC SEK and SLB management and contribution from their respective technical teams. Especially, our thanks go out to the KOC and SLB management.